



FDK™ Toolkit, Education Domain Fairness Audit

External Validation Report: LSAC Bar Passage Dataset

Second Real-World Benchmark Validation, Following Justice/COMPAS

Dataset: LSAC National Longitudinal Bar Passage Study (bar_pass_prediction.csv), 22,407 law students, USA, 1991–1997 cohort

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Abstract

This report documents the validation of the FDK™ Toolkit's Education fairness-audit pipeline against a real, publicly available benchmark dataset, the LSAC National Longitudinal Bar Passage Study, the second such real-world validation completed following the Justice domain's validation against the ProPublica COMPAS dataset. Unlike COMPAS, which required two backend fixes discovered through direct testing, the Education validation surfaced a wider range of issues spanning column-detection logic, a genuine calculation defect, a missing user-interface capability, and an important methodological pitfall in how a derived model-decision column was constructed. Four issues were identified and resolved: two keyword-detection gaps preventing real-world column names from being recognized, a data-quality defect in probability-column detection that let a partially missing binary column masquerade as a continuous probability, and the complete absence of a manual override control for the protected-attribute (group) column, which has now been added to the upload interface. A fifth methodological issue, a naive 0.5 probability threshold collapsing nearly all predictions to a single class, was identified and corrected through dataset preparation rather than a code change. Following remediation, the Education pipeline produced a complete, numerically valid 26-metric audit revealing a substantial and statistically significant racial disparity in bar-passage outcomes, consistent with this dataset's well-documented public record. This constitutes the toolkit's second successful external validation and reinforces several lessons of general applicability to the remaining five FDK™ Toolkit domains not yet validated against real data.

1. Background and Purpose

Following the Justice domain's validation against the ProPublica COMPAS dataset, the Education domain was selected as the second real-dataset validation target, using the LSAC National Longitudinal Bar Passage Study (Wightman, 1998), a dataset with a role in education and admissions fairness research comparable to COMPAS's role in criminal-justice fairness research, and likewise the subject of extensive academic and public scrutiny regarding racial disparities in outcomes.

Where the Justice validation surfaced defects specific to real-world subgroup imbalance and missing-value handling, the Education validation surfaced a broader and arguably more instructive set of issues: the toolkit's column-detection logic proved considerably less robust when confronted with a real dataset's authentic column-naming conventions (e.g. pass_bar, race1, dnn_bar_pass_prediction) than with the synthetic diagnostic datasets used throughout the toolkit's original remediation, whose column names were deliberately chosen by the toolkit's own developers.

2. Dataset Description

Property	Value
Total records	22,407 law students (22,391 after null-row cleaning)
Group column used	race1 (5 categories: white, black, hispanic, asian, other)

Property	Value
Ground-truth column	pass_bar (eventual bar exam passage, binary)
Prediction column	predicted_pass (derived; see Finding 4 below)
Probability column	dnn_bar_pass_prediction (a genuine continuous 0–1 model score, already present in the raw data)
Race group sizes (post-cleaning)	White: ~18,700 · Black: ~1,340 · Hispanic: ~1,025 · Asian: ~895 · Other: ~405 (all comfortably above the 20-sample statistical floor)

3. Findings From Real-Dataset Validation

Finding 1: Column-detection keyword gaps prevented any meaningful mapping

The first validation attempt produced a completely uninterpretable result: the pipeline mapped its ground-truth, prediction, and probability roles to gender/enrollment-status dummy columns (parttime, other, male) with no relationship to bar passage at all.

```
"column_mapping": {"group": "gender", "y_true": "parttime",
                    "y_pred": "other", "y_prob": "male"}
```

Root cause: the Education keyword lists were tuned to the vocabulary used in the toolkit's own synthetic diagnostic datasets (e.g., actual_outcome, admitted) rather than authentic real-world education terminology. Specifically, the keyword passed required an exact match to "passed" (with the -ed suffix), which does not match this dataset's pass_bar column. No keyword for "race" existed at all in the group-detection list (only the longer word "ethnicity" was present, which cannot match the shorter column names race or race1 under the toolkit's substring-matching logic). With no genuine keyword match, detection fell through to a positional fallback that selects the first remaining column of the required data type, regardless of its real meaning.

Remediation: added "race" to the group-detection keyword list. (The equivalent gap for pass_bar, adding a "pass" keyword to the ground-truth list, was identified but not implemented in code for this validation; the manual target-column override, already present in the interface, was used instead. This remains an open item; see Section 5.)

Finding 2, Probability-column detection accepted a binary column with missing values as continuous

After Finding 1's fix, the group and prediction columns resolved correctly, but the probability column was still incorrectly mapped to parttime, a binary {0,1} status flag, not a continuous probability.

Root cause, confirmed empirically: parttime contains 34 missing values. Calling .unique() on a column with missing values returns the missing marker as an additional distinct element, so a genuinely binary column with any missing data was counted as having three unique values rather than two, satisfying a "more than two values" check intended to identify continuous data. A second, independent defect compounded this: the same detection logic's fallback branch checked only col.min() >= 0 and col.max() <= 1, and pandas' min()/max() silently ignore missing values by default, so this check passes trivially for any 0/1 column regardless of the first defect.

Remediation: the uniqueness check now excludes missing values before counting, and the redundant min/max fallback branch, which could never meaningfully exclude a binary column, was removed in favor of a proper missing-value-aware range check on the non-null values only.

Field	Before Findings 1–2	After remediation
group	gender	race1 (readable labels)
y_pred	other (race dummy variable)	predicted_pass
y_prob	male / parttime (binary dummies)	dnn_bar_pass_prediction (genuine probability)

Finding 3 , No manual override existed for the group (protected-attribute) column

The upload interface already supported manually overriding the auto-detected target column, but had no equivalent control for the group column. This meant that even with Finding 1 corrected, group selection remained entirely at the mercy of column ordering and keyword coincidence: with both race (numeric codes) and race1 (readable labels) present in the raw file, whichever appeared first in file order, and whichever of "race"/"gender" happened to be encountered first during detection, determined the outcome, with no way for a user to specify their actual intent.

Remediation: a full manual-override capability for the group column was added, mirroring the existing target-column override, a new form parameter threaded from the upload route through the detection function, and a corresponding dropdown added to the upload interface, populated with every column in the uploaded file.

Finding 4: A naive prediction threshold produced a ceiling effect that masked the real disparity

The dataset's dnn_bar_pass_prediction column is a continuous probability but does not include a paired binary decision column, so one was derived by thresholding at the conventional value of 0.5. The resulting audit showed a strikingly small disparity across every group-comparison metric (statistical parity difference of approximately 0.00005; disparate impact ratio of approximately 0.9999), a result inconsistent with this dataset's well-documented literature.

Root cause, confirmed by direct inspection of the data: the model's probability estimates are overwhelmingly high across the entire population (median 0.983; 25th percentile 0.971), such that a 0.5 threshold classified 99.996% of all 22,407 students as "predicted pass", a ceiling effect leaving essentially no group any room to vary. This is not a toolkit defect; every confusion-matrix-based fairness metric is, by construction, computed from the derived binary prediction, and a saturated derived column carries no information for those metrics to detect, regardless of how real the underlying disparity is.

Remediation: the derived prediction was rebuilt using the dataset's own median probability (0.983) as the threshold rather than the conventional 0.5, restoring a genuinely balanced ~54%/46% split with real variation across groups.

Approach	predicted-positive rate	statistical_parity_difference	disparate_impact_ratio
0.5 threshold (ceiling effect)	99.996%	0.00005	0.9999
Median threshold (corrected)	53.6%	0.225	0.595

This finding is included because it is a general methodological lesson for any future real-dataset validation using a derived decision column from a continuous score: the threshold choice itself can silently determine whether a fairness audit finds anything at all, independent of the underlying pipeline's correctness.

4. Fairness Findings on the Bar Passage Dataset

With all four findings addressed, the Education pipeline produced a complete 26-metric audit across five race groups. Headline results:

Metric	Value	Interpretation
disparate_impact_ratio	0.595	Well below the conventional 0.8 four-fifths threshold; correctly flagged as a violation
statistical_parity_difference	0.225	A 22.5 percentage-point gap in predicted-positive rate between the highest and lowest group
equal_opportunity_difference	0.218	A substantial true-positive-rate gap across groups
calibration_by_group (black)	0.237	The largest calibration gap of any group, the model is least reliable for exactly the group with the lowest actual pass rate
overall_assessment	POOR	Toolkit's own composite classification

For direct comparison, the dataset's own actual (ground-truth) bar-passage rates by race, computed independently of the toolkit as a cross-check:

Race group	Actual bar-passage rate
White	96.6%
Asian	92.2%
Other	89.7%
Hispanic	87.5%
Black	77.8%

The nearly 19-percentage-point gap between the White and Black subgroups is well documented in the education and legal-education literature examining this dataset, and the toolkit's independent audit, computed via its own pipeline logic, using the corrected derived prediction column, correctly surfaces a disparity consistent with this established finding, providing external corroboration of the toolkit's correctness in a second, independent domain.

5. Scope Limitations and Open Items

- The ground-truth (`y_true`) keyword-detection gap identified in Finding 1 (`pass_bar` not matching the passed keyword) was worked around via the manual target-column override, not corrected in the detection keyword list itself. A future real dataset using similarly-abbreviated real-world terminology may still require a manual override for the target column specifically.
- The group-column dropdown added in Finding 3 has no auto-detected suggestion to pre-populate against (the shared `/api/detect-columns` preview endpoint does not currently return a suggested group), unlike the target dropdown. The user must know, and select, the correct protected-attribute column without a hint from the interface.
- This validation covers the Education domain only. Four of the seven FDK™ Toolkit domains (Business, Finance, Health, Governance, Hiring) have not yet undergone equivalent real-dataset validation.

6. Conclusion

The Education domain fairness-audit pipeline has been successfully validated against a second real, independently collected, publicly scrutinized benchmark dataset. Four genuine defects, two keyword-detection gaps, one calculation defect in probability-column detection, and one missing user-interface capability were identified, root-caused, fixed, and verified, alongside a methodological finding regarding derived-prediction thresholds that carries lessons beyond this single dataset. The pipeline's resulting fairness findings on the LSAC bar passage dataset are directionally and substantively consistent with established public research, providing external corroboration of the toolkit's correctness in a second domain following Justice/COMPAS. The Education domain is assessed as validated for real-world use, with the scope limitations in Section 5 noted for future work.

Appendix, Reference Materials

Dataset: `bar_pass_prediction.csv` (LSAC National Longitudinal Bar Passage Study, Wightman 1998), and derived working files `bar_pass_prediction_ready.csv` / `_v2.csv` (added `predicted_pass` column, corrected threshold)

Prior reports: `FDK_Toolkit_Findings_Remediation_Report.docx`; `FDK_Toolkit_Production_Readiness_Report.docx`; `Justice_COMPAS_External_Validation_Report.docx`

Modified source files: `Education/fdk_education.py` (group-detection keyword list; `y_prob` fallback missing-value handling; group-column manual override, backend and route handler); `Education/templates/upload_education.html` and `upload_education_render.html` (group-column override dropdown)

Validation reports referenced: `education_audit_report_20260721_180012.json` (0.5-threshold ceiling-effect run); `education_audit_report_20260722_095011.json` (fully resolved, final result)